Supporting Information for

Supporting Information for: Mapping the measurable determinants and predictive limits of endogenous A-to-I RNA editing

Author list pending

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Supporting Information Text

This appendix documents the complete evidence audit used to map the measurable determinants and present predictive limits of endogenous A-to-I editing, including branches that were limited, stopped, deferred, or superseded. Four evaluation regimes remain distinct throughout: *development OOF* denotes pair-disjoint cross-fitted comparisons used for discovery and selection; *selection diagnostic* denotes published-validation quantities used for checkpoints or thresholds; *canonical held-out* denotes the single evaluation after the final manifest was frozen; and *secondary* denotes robustness or descriptive analyses that do not define the benchmark claim. Evidence status is reported separately. *Selected* denotes a controlled contribution retained in the final predictor; *supported, not yet actionable* denotes a reproducible observation that did not improve the leading site predictor; *limited* denotes a small, metric-specific, or context-specific effect; and *not supported* refers only to the tested representation at the available resolution. Deferred hypotheses lacked benchmark-compatible measurements. Superseded branches were not interpreted as negative evidence.

A. Benchmark reproduction and leakage control

Immutable AdarEdit benchmark. The upstream AdarEdit reproducibility release was preserved at the immutable revision recorded in repository provenance. Environment, package, GPU, input-file, and checkpoint provenance are retained in the repository. The upstream verifier recovered 884 graph-buildable inverted-Alu pairs: 566 train, 139 validation, and 179 test, with no pair shared between partitions. Supplied Baseline GAT checkpoints reproduced the reported metrics for artery tibial, brain/cerebellum, liver, skeletal muscle, and Combined. Combined pools evidence across GTEx tissues and was never treated as a fifth tissue when constructing environmental covariates.

Each example contains both Alu arms in transcript orientation, a focal adenosine, predicted dot-bracket structure, and EditingIndex-derived label. EditingIndex $\geq 15\%$ defines edited sites, EditingIndex $< 1\%$ defines nonedited sites, and the canonical binary task excludes the intermediate interval. Nodes encode nucleotide and structure-derived features; edges represent backbone and predicted base-pair relations. The original construction used ViennaRNA for secondary structure (1).

Coordinates and authoritative observations. Genomic coordinates were reconstructed with the original strand-aware transformation from local model positions to *hg38*/GRCh38. Pair intervals came from `Pair_Alu_withStrand.bed`; raw tissue tables supplied focal positions and EditingIndex. Both `site_id = chromosome:position:strand` and `(pair_id, local_position)` were retained. Quality control required an adenosine in transcript orientation, deterministic row mapping, correct strand, and stable identity across tissues. The resulting `site_master.parquet` records labels, continuous editing values, sequence, structure, coordinates, folds, OOF predictions, and provenance.

Test quarantine, grouped folds, and inference. Canonical test paths and SHA-256 values were registered before development. The quarantine library denied protected-path access unless a single-use authorization matched the frozen manifest. Discovery used the union of published train and validation pairs divided deterministically into 5 outer folds by pair ID. Preprocessing, feature selection, dimensionality reduction, fitted residual models, checkpoints, and thresholds excluded the outer fold. A row entered an OOF table only after a model that had not seen its pair scored it.

The grouped OOF reference is an internal research instrument, not a revision of the published benchmark. Each outer model sees fewer pairs than a final fit and therefore has lower absolute performance. OOF comparisons estimate incremental value because candidate and reference share folds, training scope, and evaluation logic. The supplied-checkpoint canonical comparison answers the final benchmark question (Fig. S1). Development differences are never added arithmetically to published test scores.

We report AUROC, AUPRC, F1, precision, recall, specificity, balanced accuracy, Matthews correlation coefficient, and Brier score. Unless specified otherwise, thresholds maximized F1 on fold-internal validation data. Pair-clustered confidence intervals use 2,000 bootstrap replicates sampling pair IDs with replacement and retaining every site from a sampled pair. Effect size and clustered uncertainty, rather than site-wise P values, determine interpretation.

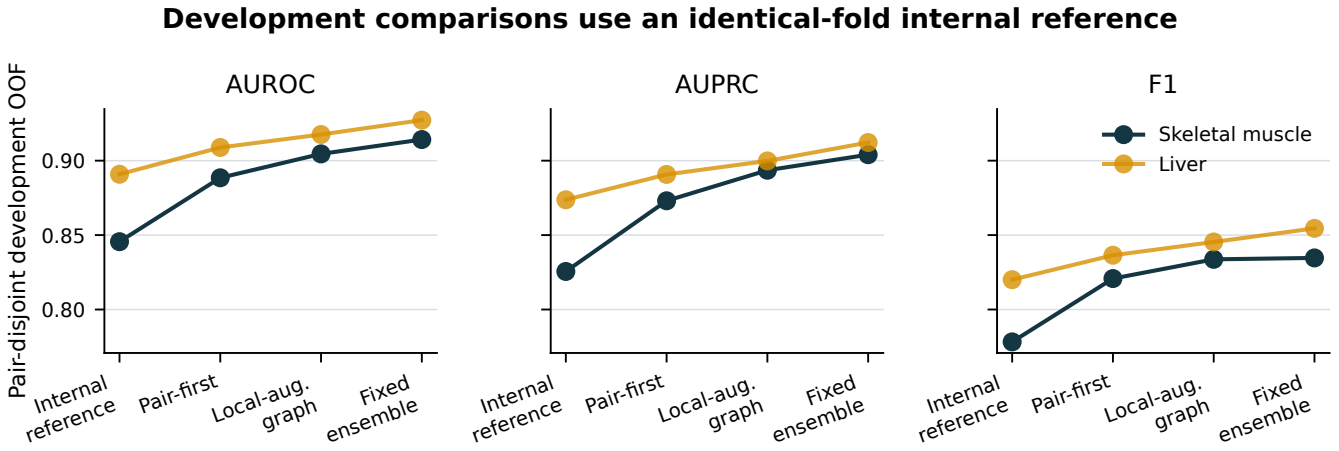


Fig. S1. Two evaluation regimes answer different questions. Development OOF comparisons use an identical-fold internal reference to attribute incremental effects. The separately reproduced supplied-checkpoint canonical test is the authoritative final benchmark.

Table S1. Complete one-use canonical held-out metrics. Thresholds were frozen from the published validation partition before test access. Combined is a pooled endpoint.

Context	Model	AUROC	AUPRC	F1	Precision	Recall	Specificity	Brier
Artery	AdarEdit	0.9238	0.9262	0.8671	0.8345	0.9023	0.7930	0.1077
Artery	ADARPro	0.9477	0.9485	0.8869	0.8840	0.8899	0.8650	0.0879
Brain	AdarEdit	0.9301	0.9286	0.8689	0.8371	0.9032	0.8021	0.1050
Brain	ADARPro	0.9508	0.9504	0.8902	0.8741	0.9069	0.8529	0.0855
Liver	AdarEdit	0.9092	0.9242	0.8541	0.8368	0.8721	0.7788	0.1196
Liver	ADARPro	0.9414	0.9500	0.8812	0.8946	0.8683	0.8670	0.0951
Skeletal muscle	AdarEdit	0.8688	0.8757	0.8139	0.7767	0.8549	0.7152	0.1526
Skeletal muscle	ADARPro	0.9144	0.9207	0.8541	0.8230	0.8876	0.7788	0.1296
Combined	AdarEdit	0.9331	0.9356	0.8674	0.8334	0.9043	0.8026	0.1043
Combined	ADARPro	0.9491	0.9513	0.8852	0.8463	0.9279	0.8159	0.0870

B. Baseline scope and transfer boundaries

Retained benchmark-characterization runs place the subsequent residual tests in context. The compact sequence–structure graph was evaluated within five human endpoints—four individual tissues and pooled Combined—and three nonhuman datasets. Within-context AUROC ranged from 0.869 to 0.933 for the human endpoints and from 0.823 to 0.865 for the nonhuman datasets. These values demonstrate substantial predictive information in sequence and predicted structure, but they are descriptive across datasets with different lineages, substrate classes, and data-generation procedures.

The full eight-by-eight transfer matrices train in one context and evaluate in every other context. AUROC is emphasized because it is not tied to transfer of a training-context decision threshold. Transfer among human endpoints was generally strong. Transfer among the three nonhuman datasets was more variable, and transfer between human and nonhuman settings was lower and asymmetric. The matrices therefore support a shared but non-universal sequence–structure grammar; they do not identify whether lineage, substrate class, assay, label distribution, or cellular state causes the transfer loss. Exact values and their source paths are retained in `results/matrix_baseline_auroc.csv` and the other metric-specific matrices.

The accompanying within-context table compares the compact baseline with a bio-aware variant. Added hand-engineered information did not improve every dataset or metric, and in some settings reduced performance. Because these runs do not share the newer OOF attribution protocol, they are used only to characterize scope and motivate the controlled campaign, not to select the final predictor. This nonmonotonic result is nevertheless an important boundary: a larger supplied feature set need not contain more usable information at the resolution of the benchmark.

C. Complete hypothesis map

Hypotheses were prioritized by mechanistic rationale, novelty relative to AdarEdit, data compatibility, leakage risk, effective sample size, and expected value. A direction advanced only after comparison with its same-fold reference and controls. Table S2

gives the quantitative synthesis; every row states its own comparator because the estimands cannot be ranked on one common scale. The machine-readable record retains historical priorities, sources, selection and stopping criteria, and final decisions. Table S10 summarizes the scientific disposition of every branch.

D. Biological-context tests

RBP occupancy, binding potential, and abundance. Candidate RBPs were prioritized independently from model outcome using known ADAR interaction, dsRNA binding, perturbation evidence, nuclear localization, Alu association, expression, and assay availability. Experimental occupancy used replicated or high-confidence ENCODE eCLIP peaks (2); predicted occupancy used PrismNet and RBPNet where compatible (3, 4). Features included focal overlap, distance to the nearest peak, local-window signal, duplex-wide burden, arm asymmetry, partner occupancy, and regulator-class aggregates. GTEx medians supplied tissue abundance and binding-by-abundance interactions (5). Combined received no artificial tissue vector.

Controls included randomized RBP identities, shuffled tissue vectors, matched-dimensional noise, abundance without binding, and binding without abundance. High-confidence baseline false positives were matched on baseline probability, tissue, structural state, and pair characteristics. None of the experimental-occupancy, predicted-binding, abundance, or interaction panels produced a reproducible residual increment beyond its controls (Fig. S2). Cell-line occupancy and tissue-average abundance are therefore unlikely to resolve regulator exposure on a particular transcript. Repeat mapping further limits eCLIP sensitivity in Alu sequence, so peak absence was never interpreted as biological absence.

Label-switching sites hold sequence and predicted structure constant across individual tissues. In the available four-tissue comparison, tissue identity nearly separated switch direction by itself, so tissue-average expression could not identify a regulator-specific transferable mechanism. Enlarging the same bulk formulation was therefore uninformative. Matched cell- or donor-level occupancy and activity measurements provide the more direct next test of established RBP regulation (6–9).

Transcript, repeat, and chromatin annotations. Reconstructed pair intervals supplied arm separation and length, orientation, repeat subfamily and divergence, Alu-neighborhood composition, transcript overlap, and exon/intron annotations. Fold-isolated residual models combined standardized features with the baseline logit. The broad site-level panel did not generalize beyond the structure graph. Section E examines geometry at the pair level.

Exact-tissue compatible H3K36me3 tracks were summarized at the target, local windows, arms, and entire pair. A chromosome-shifted overlap acted as a spatial control. H3K36me3 produced a weak liver AUPRC effect but no stable AUROC, F1, or pair-level improvement (Fig. S2). H2A.Z was not promoted to a benchmark experiment because no adequately matched and powered tissue resource was available. The weak bulk H3K36me3 effect shifts the relevant question toward quantitative, sample-matched assays of cotranscriptional state (10, 11).

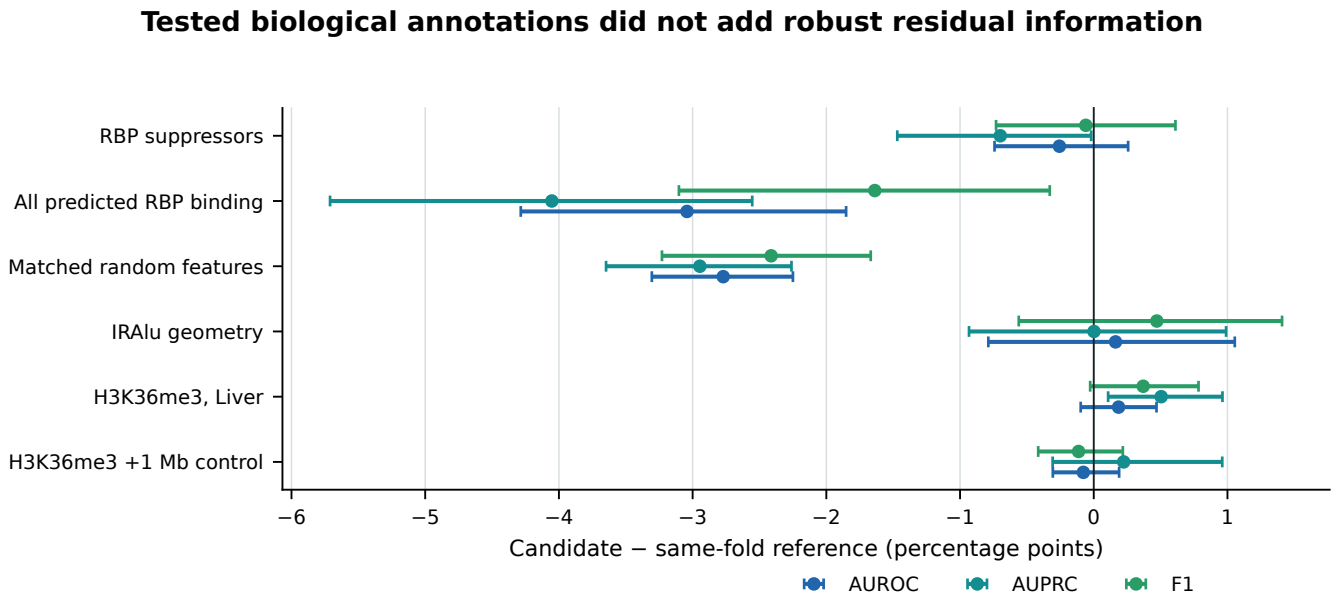


Fig. S2. Tested biological annotations and matched controls. Every point is labeled by formulation. Pair-clustered bootstrap development differences use the documented same-fold reference. The RBP, genomic, and chromatin measurements shown did not add a reproducible increment across ranking and thresholded metrics.

E. Substrate-level editing burden and realization propensity

Pair-level analyses used one row per development substrate. Outcomes were edited-site fraction, mean EditingIndex, and whether a pair contained any high-editing site. Predictors ranged from adenosine count and graph-derived sequence/structure summaries to genomic geometry and the combined 16-feature low-capacity model. All fitting remained inside pair folds.

Inter-arm gap correlated negatively with edited-site fraction in every individual tissue. Geometry plus graph summaries improved R^2 over graph summaries alone in all four tissues, with positive pair-clustered intervals (Table S3). The combined

model also classified whether a pair contained any high-editing site above chance. These observations establish reproducible substrate-level variation in editing burden. They are consistent with a realization or engagement dimension, but neither molecular process was measured directly.

Site-level translation was weaker. Multiplying or gating site probabilities with cross-fitted burden estimates increased AUPRC modestly but did not yield a stable F1 gain and worsened Brier score. A neural auxiliary engagement objective decreased the principal classification metric and met its prespecified stopping criterion. Static pair attributes can therefore describe realization propensity without recovering the time-, isoform-, transcript-, or cell-specific state required to alter the strongest site predictions.

Table S2. Development evidence synthesis. Every quantitative row states its own comparator; effects are not rankable across rows because the estimands differ. Values are percentage-point medians with 95% pair-clustered bootstrap intervals unless noted. "Not supported" refers only to the tested measurement and formulation.

Axis	Hypothesis	Comparator	Primary effect [95% CI]	Secondary behavior	Evidence status	Interpretation
Supervision	Additional valid high/low labels	Original internal reference; same folds	ΔAUROC +2.04 [+1.06, +3.00]	ΔF1 +1.42 [+0.21, +2.57]	Selected	Additional valid labels contribute independently.
Statistical unit	Pair-first weighting	Expanded row-wise sampling; identical labels	ΔAUROC +2.27 [+1.25, +3.36]	ΔF1 +2.85 [+1.68, +4.01]	Selected	Correcting substrate weight adds a second gain.
Supervision	Continuous + ordinal auxiliary labels	Pair-first binary model	ΔAUROC +0.81 [+0.10, +1.51]	ΔF1 +0.08 [-0.71, +0.86]	Limited	Ranking/calibration gain; negligible F1; adverse natural-prevalence integration.
Representation	Explicit focal-node retention	Pair-first whole-duplex pooling	ΔAUROC +1.26 [+0.52, +2.03]	ΔF1 +0.43 [-0.59, +1.45]	Limited	Ranking gain; standalone F1 interval crosses 0.
Representation	Target/contact geometry	Cross-fitted baseline logit	ΔAUROC +1.05 [+0.46, +1.61]	ΔF1 +0.19 [-0.66, +1.09]	Limited	Ranking and calibration gains localize the target effect.
Representation	Aligned target/helix-partner representation	Pair-first whole-duplex-only predictor	ΔAUROC +1.61 [+0.80, +2.38]	ΔF1 +1.31 [+0.27, +2.30]	Selected	Target-aligned fusion improves all three metrics.
Capacity control	96-nt shifted local window	Pair-first whole-duplex-only predictor	ΔAUROC -17.65 [-19.69, -15.70]	ΔF1 -11.80 [-13.37, -10.36]	Not supported	Matched capacity without target alignment is strongly adverse.
Integration	Fixed equal-logit integration	Whole-duplex-only predictor; muscle	ΔAUROC +2.56 [+2.07, +3.05]	ΔF1 +1.39 [+0.45, +2.34]	Selected	Complementary predictions; replicated in liver.
Substrate propensity	Pair geometry and editing burden	Geometry + graph summaries vs graph summaries	ΔR^2 +4.12 to +21.64 points (4 tissues)	All four pair-clustered 95% intervals exclude 0	Supported, not yet actionable	Burden is predictable; consistent with variation in realization/engagement.
Biology × AI	Cross-fitted engagement gate	Pair-first site predictor	ΔAUPRC +0.84 [+0.19, +1.47]	ΔF1 +0.41 [-0.14, +0.95]	Limited	AUPRC gain; F1 interval crosses 0; Brier worsens.
Transfer	Shared four-tissue learning	Independent tissue models; mean across 4 tissues	ΔAUROC +0.87 [+0.58, +1.15]	ΔF1 +0.75 [+0.37, +1.13]	Limited	Positive transfer below preset criteria; mean Brier worsens.
Biological context	Bulk H3K36me3 occupancy	Cross-fitted baseline logit; liver	ΔAUPRC +0.50 [+0.11, +0.96]	ΔF1 +0.37 [-0.03, +0.78]	Limited	Liver AUPRC only; no stable AUROC, F1, or pair-level gain.
Biological context	RBP binding × bulk abundance	Cross-fitted baseline logit	ΔAUROC -0.26 [-0.74, +0.26]	ΔF1 -0.06 [-0.73, +0.61]	Not supported	No residual gain; broader predicted panels were adverse.
Biological context	Transcript/repeat annotation panel	Cross-fitted baseline logit	ΔAUROC -0.41 [-1.69, +0.90]	ΔF1 -0.90 [-2.31, +0.38]	Not supported	No improvement beyond the graph representation.
Representation	Frozen RiNALMo target embedding	Recalibrated cross-fitted baseline	ΔAUROC -0.05 [-0.51, +0.39]	ΔF1 +0.70 [+0.06, +1.34]	Not supported	F1-only threshold effect; no AUROC/AUPRC gain; failed the preset rule.
Representation	Substrate-wide all-A decoder	Pair-first predictor on identical rows	ΔAUROC -6.64 [-9.25, -4.38]	ΔF1 -5.99 [-8.78, -3.65]	Not supported	Prespecified two-fold futility criterion met.
Integration	Unified end-to-end architecture	Immutable fixed ensemble	ΔAUROC -0.72 [-1.29, -0.15]	ΔF1 -0.30 [-1.25, +0.66]	Not supported	AUROC, AUPRC, and calibration were worse.
Capacity	Larger/deeper graph capacity	Supplied Bio-aware vs compact checkpoints	No common OOF effect estimate	Not uniformly better across endpoints	Not supported	No consistent advantage from generic capacity.

Table S3. Pair geometry identifies a reproducible component of substrate-level editing burden. Spearman ρ relates genomic inter-arm gap to edited-site fraction. The low-capacity pair model combines genomic geometry with graph-derived summaries; ΔR^2 is relative to graph summaries alone with a 95% pair-clustered bootstrap interval. "Any high" denotes whether a pair contains at least one site with EditingIndex $\geq 15\%$. These outcomes do not directly measure molecular realization or engagement.

Context	Gap–burden ρ	Graph-only R^2	Combined R^2	ΔR^2 [95% CI], points	Any-high AUROC
Artery	-0.482	0.053	0.270	+21.64 [+16.39, +26.93]	0.816
Brain	-0.425	0.050	0.239	+19.11 [+13.94, +23.68]	0.749
Liver	-0.475	0.070	0.277	+20.82 [+15.75, +26.03]	0.728
Skeletal muscle	-0.334	0.010	0.050	+4.12 [+0.99, +6.98]	0.705

F. Richer supervision and the statistical unit

The expanded training pool contains all eligible high- and low-editing observations in each training pair. The expanded row-wise formulation draws uniformly from those rows. The pair-first formulation draws a pair uniformly, a class with probability 0.5, and then an eligible site of that class; if the pair lacks that class, its available class is used. The canonical-row epoch budget was held fixed, preventing extra optimizer steps from explaining the difference.

The existing same-fold sequence original reference → expanded row-wise → expanded pair-first separates the two corrections (Table S4). Label recovery improved all three primary metrics. With the label pool held constant, pair-first weighting added a second favorable increment, including +2.85 F1 points. Pair-first gains replicated across all endpoints (Table S5), supporting the pair as the unit of statistical dependence.

The binary/ordinal/continuous model shared an encoder across canonical classification, ordinal bins, and transformed EditingIndex regression. It added a small ranking and Brier improvement over pair-first binary training, but essentially no F1. Adding it to the selected predictor reduced natural-prevalence F1, so the graded-label branch remained limited. Intermediate values contain ordering information, but they did not improve the primary thresholded task enough to justify added complexity.

Table S4. Same-fold decomposition of expanded supervision and pair-first weighting in muscle development data. Intervals are pair-clustered bootstrap intervals for the incremental step indicated, in percentage points.

Metric	Original reference	All high/low, row-wise	Δ from labels [95% CI]	All high/low, pair-first	Δ from pair weighting [95% CI]
AUROC	0.8456	0.8659	+2.04 [+1.06, +3.00]	0.8885	+2.27 [+1.25, +3.36]
AUPRC	0.8256	0.8520	+2.63 [+1.25, +4.17]	0.8730	+2.08 [+0.76, +3.55]
F1	0.7783	0.7924	+1.42 [+0.21, +2.57]	0.8208	+2.85 [+1.68, +4.01]

Table S5. Pair-first expanded supervision versus the same-fold internal AdarEdit reference. Values are percentage-point median differences with 95% pair-clustered bootstrap intervals.

Context	ΔAUROC [95% CI]	ΔAUPRC [95% CI]	ΔF1 [95% CI]
Artery	+1.22 [+0.80, +1.66]	+1.37 [+0.79, +1.97]	+0.69 [+0.10, +1.27]
Brain	+1.17 [+0.82, +1.54]	+1.64 [+1.02, +2.20]	+1.11 [+0.61, +1.64]
Liver	+1.79 [+1.18, +2.37]	+1.72 [+0.70, +2.71]	+1.64 [+0.96, +2.30]
Skeletal muscle	+4.28 [+3.07, +5.49]	+4.73 [+3.15, +6.35]	+4.23 [+3.11, +5.38]
Combined	+1.45 [+1.09, +1.84]	+1.69 [+1.15, +2.24]	+1.41 [+0.94, +1.98]

G. Focal and target-local representation

Three stages tested whether whole-duplex mean pooling loses site identity. First, fold-isolated target-contact geometry added AUROC and AUPRC but little F1, localizing a ranking effect near the target. Second, a focal-node graph model preserved the final target embedding alongside the global mean. It improved ranking relative to the pair model and an otherwise matched duplicate-global control, although its standalone F1 interval was insufficient for selection. Third, the selected local-augmented graph model aligned a 65-position representation of the focal adenosine and its dot-bracket helix partner. Two dilated convolutional layers encoded nucleotide, pairing, validity, target/helix-partner indicators, and relative position; the result was fused with the whole-duplex graph embedding before classification.

The local-augmented graph model exceeded the whole-duplex-only predictor in muscle and replicated in liver. Its matched-capacity control moved the same local window 96 nt away while retaining the graph, dimensions, and optimization. The

shifted model collapsed across AUROC, AUPRC, and F1 (Table S6), demonstrating that target alignment rather than generic convolutional capacity carries the improvement. Natural-prevalence results preserved the advantage (Fig. S3).

Table S6. Target-representation controls in muscle development OOF data. All values are evaluated on the same canonical development rows.

Representation	AUROC	AUPRC	F1	Brier
Whole-duplex pair model	0.8885	0.8730	0.8208	0.1372
Explicit focal node	0.9014	0.8902	0.8251	0.1342
Graph + aligned target/helix-partner	0.9045	0.8936	0.8337	0.1286
Matched-capacity +96-nt shift	0.7122	0.6943	0.7028	0.2356

H. Cross-tissue sharing

A four-output masked model shared a graph encoder across individual tissues while omitting unobserved labels from the loss. Relative to independently trained pair-first models, it yielded limited positive transfer, concentrated in the data-limited skeletal-muscle endpoint. Mean AUROC/AUPRC/F1 changes across the four individual tissue endpoints were 0.0087/0.0114/0.0075, but the predeclared AUROC and F1 selection criteria were not both met and mean Brier increased by 0.0043. Transferable intrinsic rules therefore exist, but their effect was smaller and less calibrated than the improvements from supervision and target alignment. Combined was excluded from this biological multitissue formulation.

I. Foundation-model and substrate-wide experiments

Frozen RiNALMo representations were evaluated at the target and substrate scales, including a larger frozen variant (12). Residual models and neural fusion used fold-fitted normalization and matched-dimensional controls. The 150M target view reproduced a small F1-only threshold effect, but neither variant improved AUROC/AUPRC or met the predeclared requirement to beat matched random features on both ranking and F1. The more specific next question is whether task-specific pretraining on inverted-Alu substrates would help. A corpus audit found 288,056 benchmark-excluded opposite-orientation Alu pairs in GENCODE transcript spans, but transcript connectivity, foldability, and sequence-redundancy filters were insufficiently resolved for a leakage-safe test, so that direction remains deferred.

The substrate-wide decoder encoded a pair once and classified all candidate adenosines jointly. It was evaluated against a pair-first model on identical rows with a prespecified two-fold futility criterion. The decoder lost -0.0664 AUROC and -0.0599 F1, with wholly adverse intervals, and training stopped. A relative within-pair rank feature helped a weaker original encoder but neither beat pair-first training nor its shuffled-rank control. Multiple sites provide useful supervision; the tested all-A decoder did not exploit it better than pair-first stochastic learning.

J. Natural-prevalence robustness

The canonical benchmark is balanced for direct comparison, so its precision is not a deployment-prevalence estimate. The secondary natural-prevalence evaluation retained every eligible high/low observation from unchanged development pairs. In both muscle and liver, the fixed ensemble improved ranking, AUPRC, F1, and false-positive count over its pair-first whole-duplex-only component (Fig. S3; Table S7). These results support practical false-positive reduction but do not replace the canonical task.

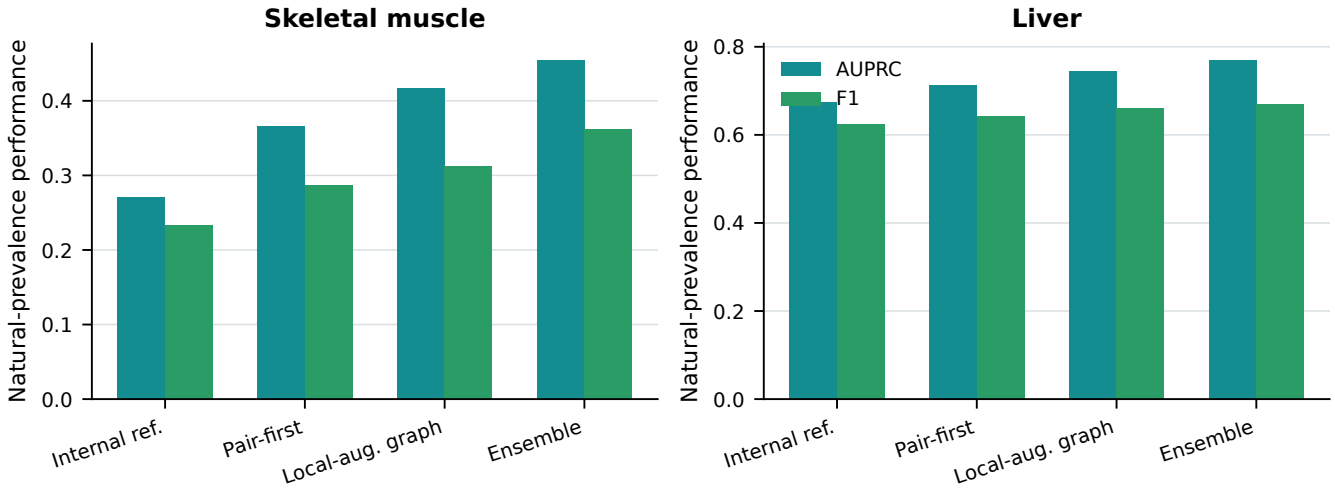


Fig. S3. Secondary natural-prevalence development evaluation. Performance for the same-fold internal reference, pair-first whole-duplex-only model, local-augmented graph model, and fixed ensemble on all eligible high/low observations from unchanged muscle and liver development pairs.

Table S7. Secondary natural-prevalence development evaluation. All eligible high/low observations from unchanged development pairs are included; fold-specific validation thresholds are used.

Context	Model	n	AUROC	AUPRC	F1	Precision	Recall	FP
Skeletal muscle	Internal reference	54289	0.8478	0.2711	0.2328	0.1351	0.8415	15804
Skeletal muscle	Pair-first whole-duplex	54289	0.8900	0.3659	0.2868	0.1714	0.8773	12441
Skeletal muscle	Local-augmented graph	54289	0.9072	0.4166	0.3125	0.1900	0.8800	11007
Skeletal muscle	Fixed ensemble	54289	0.9165	0.4540	0.3621	0.2306	0.8422	8245
Liver	Internal reference	38096	0.8915	0.6732	0.6246	0.4906	0.8592	6690
Liver	Pair-first whole-duplex	38096	0.9094	0.7128	0.6418	0.5056	0.8785	6442
Liver	Local-augmented graph	38096	0.9200	0.7432	0.6602	0.5265	0.8851	5969
Liver	Fixed ensemble	38096	0.9290	0.7687	0.6684	0.5306	0.9028	5988

K. Preregistered unified end-to-end experiment

After the prediction-level ensemble was frozen but before test access, a separate protocol specified one connected model. A shared three-layer graph attention encoder produced whole-duplex and focal-node states; the aligned local CNN supplied the third view. The three projected representations and their pairwise elementwise interactions entered a 64-unit fusion layer, while a training-only substrate head carried fixed loss weight 0.25. The connected model had 65,698 trainable parameters versus 80,138 parameters across the two independent ensemble components. Training used pair-first class balance, 1,000 epochs, Adam at learning rate 10^{-3} and weight decay 10^{-4}, batch size 256, and seed 20,260,821. The comparator prediction hash and staged decision rules were preregistered.

The unified model's AUROC and AUPRC were below the frozen ensemble in every outer fold (Fig. S4; Table S8). Aggregate F1 was similar, but ranking intervals were entirely adverse and calibration was substantially worse. The conditional no-auxiliary repair, liver replication, and all-context stages were therefore not opened. This preserves the decision rule and establishes that, in this comparison, independently trained representations were more reliably combined at prediction level than through the tested joint optimization. The experiment does not identify why optimization differed.

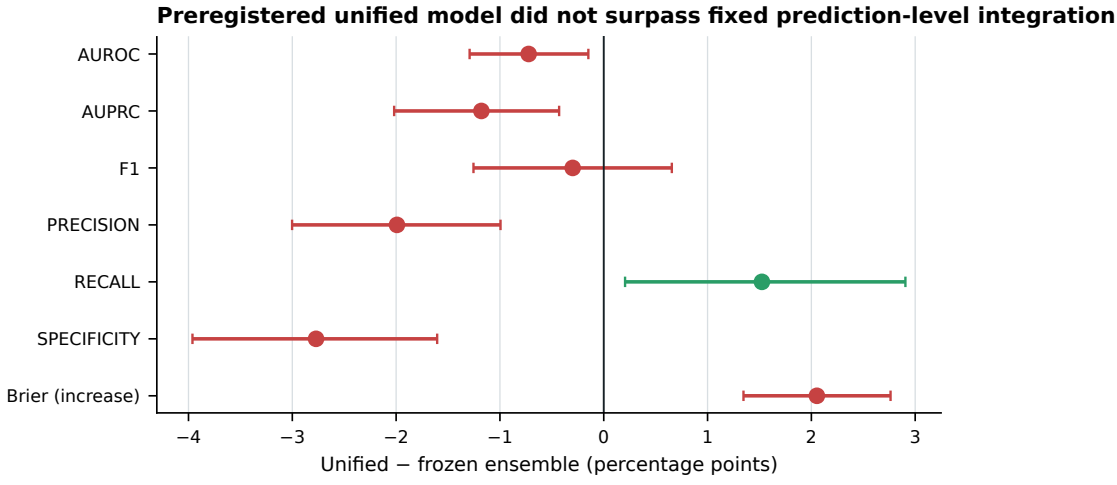


Fig. S4. Preregistered unified model versus the immutable ensemble. Bars are 95% pair-clustered bootstrap intervals for unified minus ensemble in muscle development OOF data. A positive Brier difference is adverse.

Table S8. Preregistered unified-model comparison on skeletal-muscle development data. The unified architecture was rejected before canonical-test access. Pair-clustered intervals compare the unified model with the immutable ensemble.

Model	AUROC	AUPRC	F1	Precision	Recall	Specificity	Brier
Internal reference	0.8456	0.8256	0.7783	0.7238	0.8415	0.6894	0.1695
Unified end-to-end	0.9072	0.8922	0.8317	0.8074	0.8575	0.8022	0.1372
Frozen ensemble	0.9142	0.9040	0.8347	0.8273	0.8422	0.8299	0.1168
Local-augmented graph	0.9045	0.8936	0.8337	0.7920	0.8800	0.7765	0.1286

Metric	Unified − ensemble, points [95% CI]
AUROC	-0.72 [-1.29, -0.15]
AUPRC	-1.18 [-2.02, -0.43]
F1	-0.30 [-1.25, +0.66]
PRECISION	-1.99 [-3.00, -0.99]
RECALL	+1.52 [+0.21, +2.91]
SPECIFICITY	-2.77 [-3.96, -1.60]
BRIER	+2.05 [+1.35, +2.76]

L. Final training and one-use canonical evaluation

For each endpoint, a pair-first whole-duplex-only graph model and an independently trained pair-first graph model augmented with the aligned target/helix-partner convolution were fitted on the published training pairs. Published validation data selected checkpoints and one ensemble threshold per endpoint. Ten checkpoints, histories, configurations, selected epochs, thresholds, inputs, code revision, and weights were written to `provenance/final_pipeline_manifest.json`, which retains the exact manifest SHA-256 and bound code revision. The authorization was consumed at 2026-08-25 04:28:30 UTC and records one evaluation. No subsequent model decision used test outcomes.

ADARPro improved ROC and precision–recall curves in every endpoint (Fig. S5); exact performance is in Table S1. At frozen thresholds, false-positive rescue exceeded true-positive damage. Matched-recall counts are point estimates obtained by descriptive rethresholding on final predictions; the bootstrap distribution quantifies their pair-clustered uncertainty (Table S9). Reliability curves and lower Brier scores show that ranking gains did not sacrifice calibration (Fig. S6).

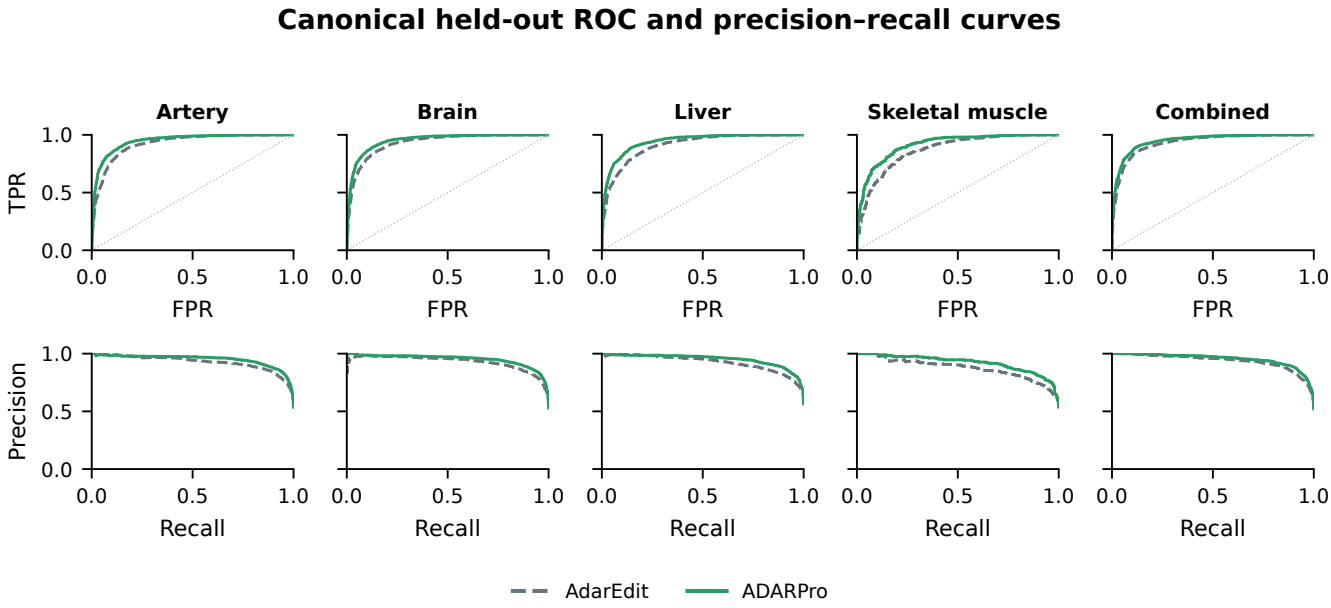


Fig. S5. Canonical held-out ROC and precision–recall curves. ADARPro and supplied-checkpoint AdarEdit are shown for four individual tissue endpoints and Combined.

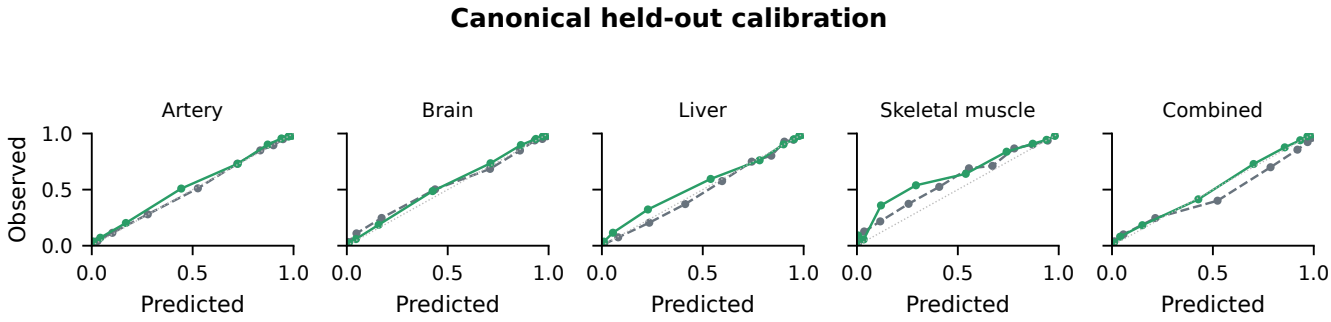


Fig. S6. Canonical held-out calibration. Reliability curves for ADARPro and supplied-checkpoint AdarEdit. Brier score is lower for ADARPro in every endpoint.

Table S9. Canonical-test false-positive transitions. Rescue and damage use frozen validation-selected thresholds. The final column is a post-test descriptive analysis that rethresholds ADARPro to match published-baseline recall exactly.

Context	Baseline FP	FP corrected (%)	New FP	TP damaged (%)	FP avoided at matched recall
Artery	460	202 (43.9%)	42	113 (4.9%)	126
Brain	425	164 (38.6%)	55	73 (3.3%)	119
Liver	399	202 (50.6%)	43	121 (5.9%)	158
Skeletal muscle	188	83 (44.1%)	41	28 (4.3%)	68
Combined	459	147 (32.0%)	116	38 (1.7%)	148

M. Complete campaign taxonomy and provenance

Table S10 records every research direction and its final scientific disposition, including branches deferred for unavailable measurements and those not pursued after stronger lower-complexity evidence emerged. The immutable registry at `experiments/experiments.csv` contains one row per run with code revision, input hashes, fold definition, seed, features, hyperparameters, artifact path, serialized metrics, and test-access status. Exact OOF predictions, fold-level tables, controls, ablations, and bootstrap results remain under `experiments/runs/`; reports 00–20 preserve chronological interpretation and stopping decisions. A machine-generated numerical claim ledger maps paper quantities to authoritative artifacts and selectors.

Table S10. Hypotheses examined and their final scientific disposition. Historical priorities, data sources, leakage risks, selection rules, stopping criteria, and result summaries remain in the machine-readable campaign record.

Axis	Hypothesis	Evidence status	Final disposition
Biology	IRAlu architecture + transcript state	Supported pair-level; site panel not supported	Broad panel terminated; geometry retained for pair-level analysis
AI	Expanded high/low supervision	Selected	Selected component
AI	Continuous + ordinal + binary multi-task	Limited	Limited ranking signal; not selected
Biology × AI	Expanded cross-tissue shared supervision	Limited	Limited shared signal; not selected
Biology × AI	Cross-fitted all-adenosine substrate profile	Not supported	Terminated after stronger-model control
Biology × AI	Substrate-wide multi-A + engagement latent	Not supported	Terminated in tested formulation
Biology × AI	Target-centered ADAR contact geometry	Limited; informed selected model	Advanced to aligned local encoder
AI	Focal-node + global graph pooling	Limited; complementary	Validated fallback/complementary view
AI	Target-local multiscale CNN + graph fusion	Selected	Selected component
Biology	Chromatin/co-transcriptional state	Limited	Bulk peak-overlap formulation terminated
Biology × AI	Hierarchical engagement × site chemistry	Limited	Stopped for futility
Biology	In vivo/ensemble RNA structure	Deferred	Deferred: coverage/context mismatch

Table continued

Axis	Hypothesis	Evidence status	Final disposition
Biology × AI	Donor/sample-level multimodal model	Deferred	Deferred: no matched donor outcomes
AI	Self-supervised IRAlu pretraining	Deferred	Deferred: corpus not benchmark-ready
AI	Frozen RNA foundation-model fusion	Not supported	Terminated after 150M-model confirmation
AI	Graph Transformer / GATv2 multi-scale encoder	Not pursued; superseded	Not pursued after target-aligned lower-capacity gain
Biology	ADAR isoform/IFN/localization state	Deferred	Deferred: no matched state/outcomes
Biology	Cell-matched perturbation RBP effects	Deferred	Deferred: matched perturbation data unavailable
AI	Fixed prediction-level equal-logit integration	Selected	Selected final integration
AI × Biology	Unified end-to-end representation learning	Not supported	Terminated after preregistered primary
AI	Larger/deeper GNN alone	Not supported	Terminated
Biology	Generic RBP binding × bulk GTEx	Not supported	Terminated

N. Interpretation and decisive future measurements

The results distinguish three biological levels and one learning principle. Whole-duplex sequence and structure encode intrinsic substrate suitability. Pair geometry and editing burden reveal a reproducible substrate-level propensity consistent with variation in realization or engagement; the underlying cellular state remains unresolved by current bulk features. Target/helix-partner alignment recovers local selectivity. Pair-first sampling respects dependence among observations from one duplex. The selected predictor combines a whole-duplex-only graph model with an independently trained graph-plus-local model and does not claim to measure the cellular environment directly.

The next decisive study should pair donor- or cell-resolved editing with transcript abundance, splicing and intron retention, ADAR1 isoform and interferon state, regulator activity, quantitative chromatin, and in vivo RNA structure from the same samples. Perturbing ILF2/ILF3/ZFR-family regulators on substrates selected for high intrinsic suitability would distinguish occupancy correlation from causal suppression. Such measurements could test a two-stage model in which cellular state controls substrate realization and local RNA context controls adenosine selectivity.

References

1. R. Lorenz, et al., ViennaRNA Package 2.0. *Algorithms for Mol. Biol.* **6**, 26 (2011).
2. E. L. Van Nostrand, et al., A large-scale binding and functional map of human RNA-binding proteins. *Nature* **583**, 711–719 (2020).
3. L. Sun, et al., Predicting dynamic cellular protein–RNA interactions by deep learning using in vivo RNA structures. *Cell Res.* **31**, 495–516 (2021).
4. M. Horlacher, et al., Towards in silico CLIP-seq: predicting protein–RNA interaction via sequence-to-signal learning. *Genome Biol.* **24**, 180 (2023).
5. GTEx Consortium, The GTEx Consortium atlas of genetic regulatory effects across human tissues. *Science* **369**, 1318–1330 (2020).
6. E. C. Freund, et al., Unbiased identification of trans regulators of ADAR and A-to-I RNA editing. *Cell Reports* **31**, 107656 (2020).
7. G. Quinones-Valdez, et al., Regulation of RNA editing by RNA-binding proteins in human cells. *Commun. Biol.* **2**, 19 (2019).
8. E. J. Park, et al., Primate lineage specification requires suppression of Alu hyperediting (bioRxiv preprint) (2026).
9. Y. Yi, et al., A dual role of EZH2 in regulating A-to-I RNA editing and mRNA stability through ADAR. *Nat. Commun.* **17**, 4421 (2026).
10. P. Cai, et al., Chromatin state shapes site-specific A-to-I RNA editing (bioRxiv preprint) (2026).
11. ENCODE Project Consortium, Expanded encyclopaedias of DNA elements in the human and mouse genomes. *Nature* **583**, 699–710 (2020).
12. R. J. Penic, T. Vlasic, R. G. Huber, Y. Wan, M. Sikic, RiNALMo: general-purpose RNA language models can generalize well on structure prediction tasks. *Nat. Commun.* **16**, 5671 (2025).